

Disproof of conjecture 00000000437

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Verdict: FALSE

We disprove conjecture 00000000437 as stated. The numerical formula in the conjecture (Witt's formula) is a true classical theorem, but the appended *parity clause* is false. The refutation is unconditional and elementary: two powers of two, namely $n = 1$ and $n = 2$, have different parities, $L_1 = 2$ and $L_2 = 1$, so the parity of L_n is not determined by whether n is a power of two.

1 The conjecture

Quoted verbatim from `conjectures/00000000437.md`:

Definition: The shuffle algebra $\text{Sh}(V)$ (tensor algebra with shuffle product). **Conjecture:** The n -th graded dimension of the Lie-primitive space of $\text{Sh}(V)$ for $\dim V = 2$ is $\frac{1}{n} \sum_{d|n} \mu(d) \cdot 2^{n/d}$ (Witt's formula), and its parity is determined explicitly by whether n is a power of 2.

中文。定义: shuffle 代数 $\text{Sh}(V)$ (张量代数带 shuffle 积)。猜想: shuffle 代数 $\text{Sh}(V)$ 的李代数本原空间的第 n 个分次维数在 $\dim V = 2$ 时为 $\frac{1}{n} \sum_{d|n} \mu(d) \cdot 2^{n/d}$ (Witt 公式), 其奇偶性由 n 是否为 2 的幂显式给出。

For the rest of the document let

$$L_n = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d}, \quad (1)$$

where μ is the Möbius function. Thus L_n is the number of binary Lyndon words of length n , equivalently the number of aperiodic binary necklaces of length n , equivalently the number of monic irreducible polynomials of degree n over $\mathbb{F}_2$; all of these satisfy (1) (Witt's formula). That formula is *not* in dispute. The conjecture consists of two claims, and only the second is false:

1. **(Witt's formula, true.)** The n -th graded dimension is L_n .

2. (**Parity clause, false.**) The parity of L_n “is determined explicitly by whether n is a power of 2.”

2 The value table

The values of L_n and their parities for $n \leq 20$ are as follows. The last column records the predicate “ n is a power of two” (with the standard convention $2^0 = 1$, so $n = 1$ is a power of two).

n	L_n	parity of L_n	n is a power of 2
1	2	even	yes
2	1	odd	yes
3	2	even	no
4	3	odd	yes
5	6	even	no
6	9	odd	no
7	18	even	no
8	30	even	yes
9	56	even	no
10	99	odd	no
11	186	even	no
12	335	odd	no
13	630	even	no
14	1161	odd	no
15	2182	even	no
16	4080	even	yes
17	7710	even	no
18	14532	even	no
19	27594	even	no
20	52377	odd	no

Reading the table, the parity of L_n visibly fails to track the power-of-two predicate: $n = 1, 2$ are both powers of two yet the parities are even and odd respectively; $n = 8, 16$ are powers of two with even parity; and $n = 6, 10, 12, 14$ are not powers of two yet have odd parity.

3 The refutation

3.1 The sharpest witness: $n = 1$ and $n = 2$

Theorem 1. $L_1 = 2$ and $L_2 = 1$. Both $1 = 2^0$ and $2 = 2^1$ are powers of two, but L_1 is even and L_2 is odd. Hence there is no well-defined function of the predicate “ n is a power of two” that returns the parity of L_n ; the parity clause of conjecture 00000000437 is false.

証明. Directly from (1). For $n = 1$ the only divisor is $d = 1$ and $\mu(1) = 1$, so $L_1 = 1 \cdot 2^1/1 = 2$. For $n = 2$ the divisors are $d = 1, 2$ with $\mu(1) = 1$, $\mu(2) = -1$, so

$$L_2 = \frac{\mu(1)2^2 + \mu(2)2^1}{2} = \frac{4 - 2}{2} = 1.$$

Thus $L_1 = 2 \equiv 0 \pmod{2}$ while $L_2 = 1 \equiv 1 \pmod{2}$. Since $1 = 2^0$ and $2 = 2^1$ are both powers of two, the two arguments that the parity clause says determine the same value in fact give different values. $\square$

Remark 2 (Under every faithful reading). Read the clause as a statement about the truth value of the predicate $P(n) := “n \text{ is a power of two}”$. Then $P(1) = P(2) = \text{true}$ while $L_1 \bmod 2 \neq L_2 \bmod 2$, so no function of $P(n)$ can equal $L_n \bmod 2$. The failure does not depend on which explicit formula the file had in mind: the file supplies none, and none can exist, because the parity is not a function of $P(n)$.

3.2 The fallback witness: $n = 2$ and $n = 8$

If one declines to count 1 as a power of two (some authors exclude 2^0), the refutation is unchanged, because the two *proper* powers of two 2 and 8 already have different parities:

$$L_2 = 1 \equiv 1 \pmod{2}, \quad L_8 = \frac{2^8 - 2^4}{8} = \frac{256 - 16}{8} = 30 \equiv 0 \pmod{2}.$$

So even under the restricted convention “powers of two are 2, 4, 8, 16, ...”, parity is not determined by whether n is a power of two.

3.3 The “if and only if” reading fails at $n = 6$

A natural sharpening of the clause is the equivalence

$$“L_n \text{ is odd} \quad \text{iff} \quad n \text{ is a power of two}”.$$

This is false in *both* directions.

- $n = 8$ is a power of two, but $L_8 = 30$ is even. ($n = 16$ likewise: $L_{16} = 4080$ is even.)

- $n = 6$ is not a power of two, yet

$$L_6 = \frac{\mu(1)2^6 + \mu(2)2^3 + \mu(3)2^2 + \mu(6)2^1}{6} = \frac{64 - 8 - 4 + 2}{6} = \frac{54}{6} = 9$$

is odd. The obstruction is intrinsic: $6 = 2^1 \cdot 3$ has $v_2(6) = 1$ and odd part 3, which is squarefree.

So neither implication holds, and the parity clause is false however the unspecified “explicit formula” is read.

4 The true repaired characterisation

Although parity is *not* governed by the power-of-two predicate alone, it does have an exact description. It was verified numerically for all $n \leq 60$ by `reproduce.py`; here is the proof.

Proposition 3. *Write $n = 2^a m$ with m odd and $a = v_2(n) \geq 0$. Then*

$$L_n \text{ is odd} \iff a \in \{1, 2\} \text{ and } m \text{ is squarefree.}$$

Equivalently: L_n is even for every odd n , and for even n the value L_n is odd exactly when $v_2(n) \in \{1, 2\}$ and the odd part of n has no repeated prime factor.

证明. Multiplying (1) by n gives the integer identity

$$n L_n = \sum_{d|n} \mu(d) 2^{n/d},$$

whose nonzero summands are indexed by the squarefree divisors d of n . On that set the exponent n/d is strictly decreasing in d , so there is a unique squarefree divisor realising the smallest exponent, namely the largest one, $d^* = \text{rad}(n) = \prod_{p|n} p$. Every other summand is divisible by $2^{n/d^*+1}$, hence does not cancel the lowest-order term; consequently

$$v_2(n L_n) = \frac{n}{\text{rad}(n)}.$$

Since $v_2(n) = a$ and $v_2(n L_n) = a + v_2(L_n)$, this gives

$$v_2(L_n) = \frac{n}{\text{rad}(n)} - a.$$

If $a = 0$ (that is, $n = m$ is odd) then $\text{rad}(n) = \text{rad}(m)$ and $v_2(L_n) = m/\text{rad}(m) \geq 1$, so L_n is even.

If $a \geq 1$ then $\text{rad}(n) = 2 \text{rad}(m)$ and

$$v_2(L_n) = 2^{a-1} \frac{m}{\text{rad}(m)} - a.$$

This vanishes exactly when (i) $m = \text{rad}(m)$, i.e. m is squarefree, and (ii) $2^{a-1} = a$, i.e. $a \in \{1, 2\}$: for $a \geq 3$ one has $2^{a-1} > a$, so the expression is positive, and for $a \in \{1, 2\}$ it equals $m/\text{rad}(m) - a$, which is 0 precisely when m is squarefree. Hence L_n is odd exactly for $a \in \{1, 2\}$ with m squarefree. $\square$

Corollary 4. *The parity is governed by $v_2(n)$ and the squarefreeness of the odd part of n together. The power-of-two predicate alone is insufficient: among the powers of two, L_1, L_2, L_4 are odd (for $n = 1, 2, 4$ with odd part 1 squarefree and $v_2(n) \in \{0, 1, 2\}$) while L_8, L_{16}, L_{32} are even ($v_2(n) \geq 3$). This is exactly the failure recorded in Theorem 1.*

5 Caveats

1. **Witt’s formula is true and is not being disproved.** The numerical part of the conjecture, $L_n = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d}$, is a standard theorem (binary Lyndon words / aperiodic necklaces / irreducible polynomials over $\mathbb{F}_2$). The verdict FALSE refers solely to the parity clause.
2. **The parity clause is under-specified.** The file gives no explicit formula for the parity, only the assertion that it “is determined explicitly by whether n is a power of 2.” The refutation is robust: the sharpest witness (1, 2), the fallback (2, 8), and the iff-violation ($n = 6$) defeat, respectively, the functional reading, the restricted-convention reading, and the iff reading. No reading in which the parity is a function of the power-of-two predicate alone can survive.
3. **Convention on 1.** Under the convention $2^0 = 1$, the pair (1, 2) is the smallest witness. Under the convention that 1 is not a power of two, the pair (2, 8) is the smallest witness. The verdict is the same.
4. **The repaired statement changes the hypothesis.** Replacing “ n is a power of two” by the pair of conditions in Proposition 3 yields a correct criterion; it is not a cosmetic rewording of the conjecture, since $v_2(n) \in \{1, 2\}$ and squarefreeness of the odd part are independent of the power-of-two predicate. For instance $n = 6$ has odd L_6 yet is not a power of two, and $n = 8$ is a power of two with even L_8 .
5. **Scope of the numerics.** The values L_n for $n \leq 60$ and the repaired characterisation are checked by `reproduce.py`; the finite witnesses $L_1, L_2, L_6, L_8, L_{24}$ and the parity statements are proved in core Lean in `lean4/Main.lean`. The value table up to $n = 20$ is quoted in Section 3.

6 Reproducibility and formalisation

The arithmetic is reproduced by `reproduce.py` (Python 3 standard library only), which computes μ by factorisation, evaluates L_n for $n \leq 60$, prints the table, asserts $L_1 = 2$, $L_2 = 1$, $L_6 = 9$, that 1 and 2 are powers of two with different parities, the (2, 8) fallback, the $n = 6$ iff-violation, and the repaired characterisation; it prints **PASS** and exits 0 iff every check holds.

The finite witnesses are formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). It proves, among others,

```
theorem witt_1 : witt 1 = 2
theorem witt_2 : witt 2 = 1
theorem witt_6 : witt 6 = 9
theorem witt_8 : witt 8 = 30
theorem isPow2_1 : isPow2 1 = true
theorem isPow2_2 : isPow2 2 = true
theorem isPow2_6 : isPow2 6 = false
theorem witness_1_2_parity : witt 1 % 2 ≠ witt 2 % 2
theorem fallback_2_8_parity : witt 2 % 2 ≠ witt 8 % 2
theorem parity_not_determined_by_pow2 :
  ¬ (∀ n m, isPow2 n = isPow2 m → witt n % 2 = witt m % 2)
theorem conjecture_00000000437_false : (the collected refutation)
```

The Möbius function is computed from an *ascending* smallest-prime-factor search; this ordering is essential, since a “largest divisor $\leq \sqrt{N}$ ” search returns the composite 4 for $N = 20$ and $N = 24$ and silently corrupts μ (for instance producing $L_{24} = 698869$ instead of the correct $L_{24} = 698870$, proved as `witt_24`). Every finite fact is closed by `decide`, never `native_decide`, so that no `Lean.ofReduceBool` enters the axiom footprint; the `#print axioms` audit in `Check.lean` reports only `propext` (and no axioms at all for the purely computational facts).

7 Files

File	Description
README.md	Verdict, bilingual quote, tables, witnesses, caveats, status.
main.tex	This document. Standalone <code>article</code> ; <code>tectonic main.tex</code> .
build/main.pdf	PDF produced by <code>tectonic main.tex</code> .
reproduce.py	Standard-library reproduction; <code>PASS/FAIL</code> , non-zero on failure.
lean4/lean-toolchain	Pins <code>leanprover/lean4:v4.33.1</code> .
lean4/lakefile.toml	Lake configuration; library <code>Main</code> , package <code>tlmc437</code> .
lean4/Main.lean	Core-Lean formalisation of the witnesses.
lean4/Check.lean	<code>#print axioms</code> audit.
lean4/README.md	Statement table and scope note.